AI ACADEMY · PYTHON READINESS BRIDGE · STUDENT LESSON

Expressions and operator order

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Python readiness portfolio

THE CONNECTED WEEK

State the model. Mark the grouping. Check the domain.

Evaluate expressions while distinguishing ordinary division, whole groups and remainder.

Keep one habit: Packing model → dimensionless parentheses → five grouping/boundary cases → selected transfer → evidence sheet.

MAIN LESSON · SECTION 1

Which quantities belong inside each group?

> Express the intended rule before checking the arithmetic.

Your learning goals

I can evaluate numeric **expression**s using Python's grouping rules.

I can use **parentheses** to express the intended calculation.

I can distinguish ordinary division, whole groups and a **remainder**.

Recall before notes

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.
- Without opening the old lesson, explain one key idea from Values variables and types. Give an example and a mistake that the idea helps you avoid.
- After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.
- This week you will practise:
- Evaluate numeric **expression**s using Python's grouping rules
- Use **parentheses** to express the intended calculation
- Distinguish ordinary division, whole groups and a **remainder**
- What is the difference between 8 + 2 and "8" + "2"?

My starting explanation

Your end goal

Trace **expression**s and **operator** order and explain the changed program before running it.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

Retrieve what you already know

This week’s end goal

MAIN LESSON · SECTION 2

Three packs of eight, plus two loose counters

> The same numbers can express different situations when grouped differently.

Read the physical counting contract

Three packs each contain eight counters, with two more loose counters. First count the packed counters, then add the loose ones. A program needs that structure, not just the right numbers.

This week you will predict **expression**s, make grouping visible and check division against a physical interpretation.

Use a saved numeric-expression script

Use numeric values in a fresh Python 3 script called **expression**s.py. Keep expected and observed results separate as before. Multiplication and division group more tightly than addition and subtraction.

Parentheses let you state a different grouping. A correct calculation still depends on correct quantities and units.

Your first explanation

List the given facts and one fact that is not supplied.

MAIN LESSON · SECTION 3

Match parentheses to the written model

> Adding a loose pile once differs from adding it to every pack.

Compare three explicit groupings

Two expressions with different meanings

EXPRESSION	FIRST CALCULATION	FINAL VALUE
3 * 8 + 2	3 * 8 is 24	24 + 2 is 26
3 * (8 + 2)	8 + 2 is 10	3 * 10 is 30
(3 * 8) + 2	3 * 8 is 24	24 + 2 is 26

Adding two inside every pack differs from adding two loose counters once.

Complete pack-and-loose script

```
packs = 3
per_pack = 8
loose = 2
total = packs * per_pack + loose
print(total)
```

Explain why the model gives 26 rather than 30

The script displays 26. Replacing the total **expression** with packs * (per_pack + loose) displays 30, because it adds two to every pack. **Parentheses** are not decoration: they can change the represented situation.

Write a brief phrase for each group before running.

Explain the deciding step

- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Compare expression results without changing their meaning silently

> State which quantities change and which grouping remains fixed.

Predict the supported change

- Predict 16 - 5 - 3, 16 - (5 - 3), and 18 / 3 * 2. Write the grouping, run each line and record actual results separately.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the pictured variant

- Change: Replace the final multiplier 5 with 4 in both **expression**s.

▼ **Check after predicting**

Prediction: The first result is 11; the grouped result is 20.

Reason: 3+2*4=3+8; (3+2)*4=5*4.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Trace precedence in a separate dimensionless example

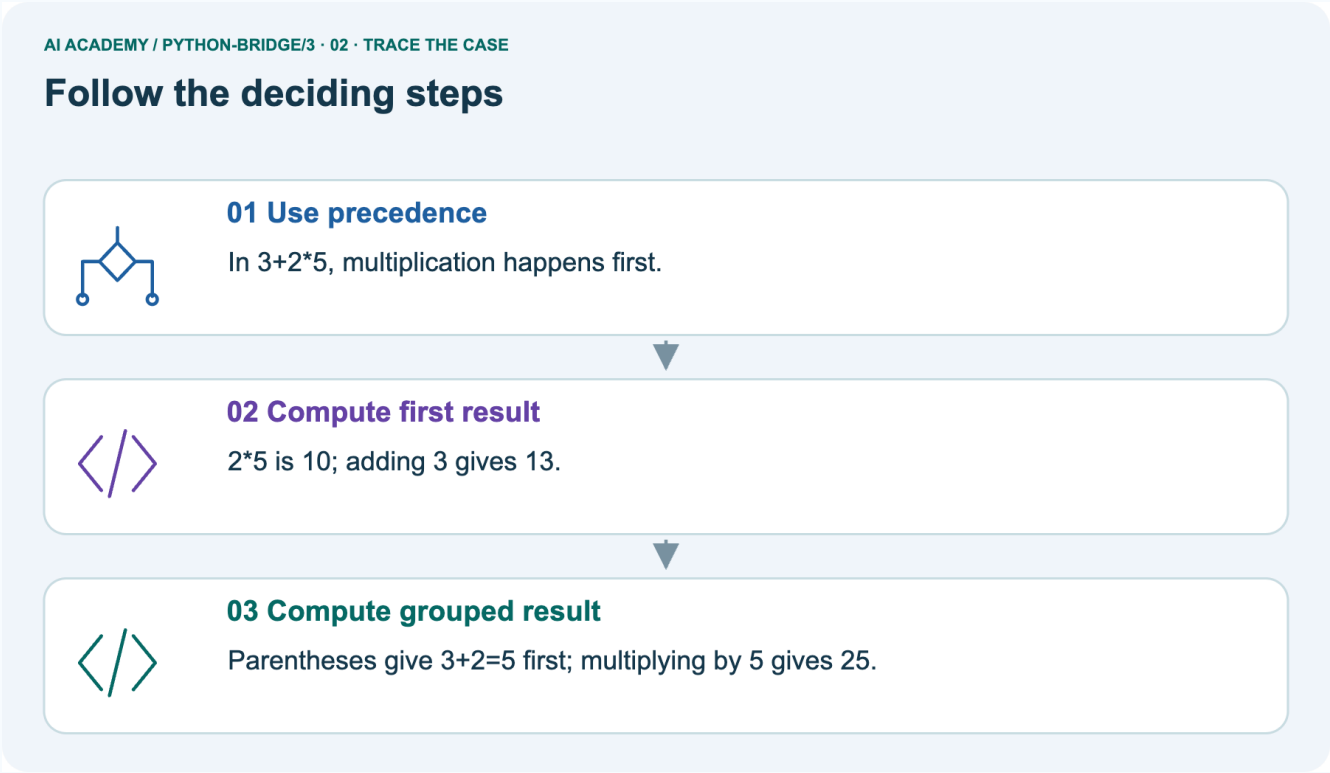
> Do not mix the 3+2*5 picture with the 3*8+2 packing story.

Read the supplied expression pair

Compare two dimensionless arithmetic **expression**s: adding 3 after 2*5 versus multiplying the whole 3+2 group by 5. Match each group to a written model before using it for a real counting task.

Python 3: print(3+2*5); print((3+2)*5). **Input**s are dimensionless for this arithmetic exercise.

Trace each deciding step



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Use precedence	In 3+2*5, multiplication happens first.
2. Compute first result	2*5 is 10; adding 3 gives 13.
3. Compute grouped result	Parentheses give 3+2=5 first; multiplying by 5 gives 25.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Parentheses explicitly group an operation. Without them, Python multiplication takes priority over addition; the same numerals can therefore describe different procedures.

Read the complete comparison program

```
print(3 + 2 * 5)
print((3 + 2) * 5)
```

Predict changing the multiplier to 4

- Change one thing: Replace the final multiplier 5 with 4 in both **expression**s. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting

AI ACADEMY / PYTHON-BRIDGE/3 · 04 · COMPARE THE CASES

Change one thing. Trace what follows.

Original case

Parentheses give 3+2=5 first; multiplying by 5 gives 25.

Changed condition

Replace the final multiplier 5 with 4 in both expressions.

Worked changed-case result

The first result is 11; the grouped result is 20.
Why: 3+2*4=3+8; (3+2)*4=5*4.

Follow the labelled picture with the matching explanation. Open the vector diagram

State the model-choice limit

- Arithmetic correctness does not decide which **expression** matches a real problem; the written model must define the intended quantities.

- Read the labelled picture: Trace the operation order in both **expression**s. Identify which picture step changes with **parentheses** and predict both results when the multiplier becomes four.

▼ Separate practice · Same-precedence operations

```
print(20 - 6 - 2)
print(20 - (6 - 2))
print(12 / 3 * 2)
```

The first line groups subtraction from left to right: (20 - 6) - 2 gives 12. The second gives 16. Multiplication and division share a **precedence** level, so 12 / 3 * 2 groups as (12 / 3) * 2 and displays 8.0, not 2.0.

Ordinary / division produces a float, even when the result is mathematically whole.

> **Explain why 18 - 4 * 3 gives 6 while (18 - 4) * 3 gives 42. Which operation is grouped first in each expression?**

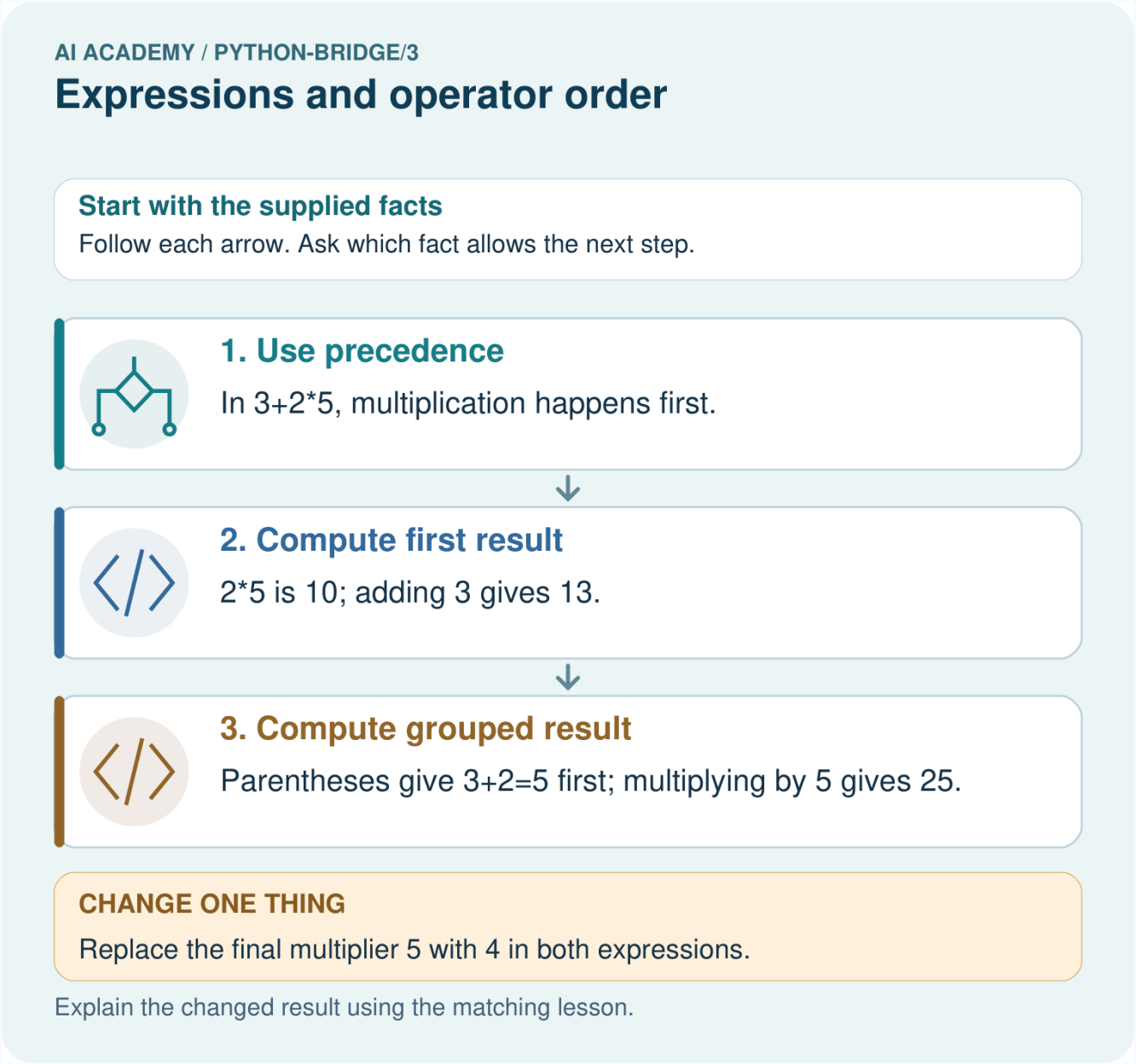
▼ **Earlier wording, alternatives and picture transcripts**

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Expressions and operator order

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Choose division, complete groups or a remainder

> The counting interpretation assumes nonnegative whole counters and a positive group size.

Run the complete 17-in-groups-of-5 script

```
counters = 17
group_size = 5
print(counters / group_size)
print(counters // group_size)
print(counters % group_size)
```

Check 17=3*5+2 and the three output meanings

The **output**s are 3.4, 3 and 2. With nonnegative whole counters and a positive group size, // gives the number of complete groups and % gives the **remainder**: 17 = 3 * 5 + 2.

Ordinary division answers a different question about a ratio; 3.4 complete five-counter groups cannot be built without splitting counters.

State the zero/negative domain boundary

For nonnegative whole **input**s, the **remainder** is at least zero and smaller than the group size. A group size of zero is invalid and division by zero raises an error.

These whole-group interpretations assume nonnegative counts and positive group sizes; do not generalise them to negative quantities without examining Python's floor-division rule.

Explain the operation choice

- For 17 counters in groups of 5, explain /, // and %. Show a multiplication-and-addition check for the complete groups and **remainder**.

State one repair and the checks you would rerun.

▼ **Earlier wording, alternatives and picture transcripts**

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise grouping with less help

> Mark parentheses before calculating or running.

Explain the procedure

- Write an **expression** for three packs of eight counters plus two loose counters. Explain why adding two within the **parentheses** for every pack describes a different situation.

Explain why 18 - 4 * 3 gives 6 while (18 - 4) * 3 gives 42. Which operation is grouped first in each expression?

> **Explain your prediction to a partner before comparing results.**

Complete the W2 grouping task

Use workbook W2 for the connected example "**Expression**s and **operator** order". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Audit left-to-right reasoning

- Use the worked case for **Expression**s and **operator** order. Proposed explanation: "All operations are read left to right regardless of **precedence**". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the pictured multiplier change

- Trace the operation order in both **expression**s. Identify which picture step changes with **parentheses** and predict both results when the multiplier becomes four.

My explanation using a pictured step and the given facts

Reserve the selected transfer

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Predict then test expressions

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Test five complete grouping and division cases

> Write the grouping and domain before observing the output.

All supplied expression and boundary cases

Complete case	Expected grouping and result
A · 16-5-3	(16-5)-3=8

Complete case	Expected grouping and result
B · 16-(5-3)	16-2=14
C · 18/3*2	(18/3)*2=12.0
D · 17 counters/group_size 5	3.4 ratio, 3 whole groups, 2 remainder; 17=3*5+2
E · 17 counters/group_size 0	Division, floor division and modulo each raise ZeroDivisionError; no valid grouping count

Complete W2 program

```
print(16 - 5 - 3)
print(16 - (5 - 3))
print(18 / 3 * 2)
```

Predict, run or trace, then explain meaning

1. Predict A/B/C from the complete W2 program before running, then record actual output separately. D uses the supplied division program; explain each result's task meaning. Test E one operation at a time in a separate script, because its error stops subsequent statements. Keep the positive-group-size contract explicit; a zero denominator is not a group count of zero.

   Use the printed equality to check quotient and remainder for valid nonnegative integer cases. A changed expression may run correctly yet represent the wrong physical rule. Keep the selected grouped-addition/floor-division task for an individual attempt; record prior exposure to the familiar 17/5 example.

Keep the evidence record

Predict 16 - 5 - 3, 16 - (5 - 3), and 18 / 3 * 2. Write the grouping, run each line and record actual results separately.

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Reference · Original complete scripts

```
packs = 3
per_pack = 8
loose = 2
total = packs * per_pack + loose
print(total)
```

```
print(20 - 6 - 2)
print(20 - (6 - 2))
print(12 / 3 * 2)
```

```
counters = 17
group_size = 5
print(counters / group_size)
print(counters // group_size)
print(counters % group_size)
```

▼ Earlier wording, alternatives and picture transcripts

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: compare the selected grouped expressions

> Use the stated independent code before feedback.

Selected complete task and domain

Predict each line. Explain which calculation groups addition before division. What would (17+5)%5 return, and how do /, // and % differ for the supplied positive integers?

```
print((17 + 5) // 5)
print(17 + 5 // 5)
print(17 / 5, 17 // 5, 17 % 5)
```

Trace each line from its exact parentheses and operators. The final division facts may be familiar from supported practice; explain their meaning rather than treating a remembered number as independent evidence. Use a fresh parallel input after support or prior exposure, keeping nonnegative counts and positive divisors for the grouping interpretation.

Your response and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Additional practice · Earlier division transfer

Four trays hold six tiles each and five tiles are loose. Write and evaluate the total **expression**. Then pack the total into complete groups of four and find the **remainder**. Explain why 4 * (6 + 5) is wrong for this story.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Group an expression deliberately

Attempt this without hints

Use feedback to improve

MAIN LESSON · SECTION 11

Keep an expression-and-model evidence sheet

> The rule's meaning and Python's result belong in separate fields.

This week's contribution

Add a labelled **expression**s and **operator** order **evidence** sheet to Python readiness portfolio. Show how you can evaluate numeric **expression**s using python's grouping rules.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project checkpoint and evidence

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Finish: explain grouping, operation and input assumptions

> Correct arithmetic does not choose the physical model for you.

Explain without notes

- Close your notes. Explain this goal in your own words: Evaluate numeric **expression**s using Python's grouping rules

> **My deciding step, evidence and one remaining question**
>
> _____
>
> _____
>
> _____
>
> _____

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · The words used in this procedure**

REFERENCE · SECTION 5

Reference · The words used in this procedure

> Explain each term using a particular code line.

Vocabulary and response

Word	Meaning
Expression	Code evaluated to produce a value
Operator	A symbol such as + or * that specifies an operation
Precedence	Rules that determine how an expression is grouped
Parentheses	Symbols that make a grouping explicit
Remainder	The amount left after forming whole groups

Use two terms to explain a step in this week's example.

REFERENCE · SECTION 10

Extra challenge · Explain, apply and challenge

> Choose the next useful task after supported practice.

Choose a challenge

- Write an **expression** for three packs of eight counters plus two loose counters. Explain why adding two within the **parentheses** for every pack describes a different situation.
- For 17 counters in groups of 5, explain /, // and %. Show a multiplication-and-addition check for the complete groups and **remainder**.
- Explain why a group size of zero must be excluded. State the assumptions that let complete groups and a **remainder** describe the packing task.

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Earlier wording, alternatives and picture transcripts

Explain → apply → challenge

▼ Reference · Inspect your evidence

REFERENCE · SECTION 12

Reference · Inspect your evidence

> Use the record after practice rather than revealing the independent answer.

Self-check and explanation

- Keep the **expression**, substituted numbers, grouping, units and actual result. A bare number does not show why it answers the question.

  If an edit changes the grouping, preserve the old result and rerun a case where the **expression**s differ. Week 4 will turn comparisons into Boolean conditions.

- Explain why a group size of zero must be excluded. State the assumptions that let complete groups and a **remainder** describe the packing task.

> **A team has 23 whole tiles and puts 4 in each complete tray. Which operations find complete trays and leftover tiles? Predict both values and verify them by reconstructing 23. Explain why / alone does not answer both questions.**

My limited claim and a useful next check

▼ **Extra practice · A complete trace**

REFERENCE · SECTION 13

Extra practice · A complete trace

> Predict before opening the supplied trace.

Explain the input and procedure

- Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ Check the complete trace

Two expressions with different meanings

EXPRESSION	FIRST CALCULATION	FINAL VALUE
3 * 8 + 2	3 * 8 is 24	24 + 2 is 26
3 * (8 + 2)	8 + 2 is 10	3 * 10 is 30
(3 * 8) + 2	3 * 8 is 24	24 + 2 is 26

Adding two inside every pack differs from adding two loose counters once.

▼ Earlier wording, alternatives and picture transcripts

Complete evidence record

▼ Reference · Earlier route checklist

REFERENCE · SECTION 15

Reference · Earlier route checklist

> Keep one main lesson route and use this as a reminder.

Checklist and first prediction

By the end, I can	How I will show it
Evaluate numeric expressions using Python's grouping rules	Explain the deciding step, show a trace or test, and state one limit.
Use parentheses to express the intended calculation	Explain the deciding step, show a trace or test, and state one limit.
Distinguish ordinary division, whole groups and a remainder	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: **Parentheses** change an **expression**. A toy shelf calculator compares adding a fixed cost after multiplication with multiplying an entire group.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 17

Extra practice · Predict, check and repair

> Use only after an attempt; keep the reserved assessment separate.

First question and hint

- **Parentheses** change an **expression**. A toy shelf calculator compares adding a fixed cost after multiplication with multiplying an entire group.
- Which labelled step matches this detail? **Parentheses** give 3+2=5 first; multiplying by 5 gives 25.
- 1. Compute grouped result
- 2. Use **precedence**
- 3. Compute first result
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> _____
>
> _____
>
> _____
>
> _____

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Compute grouped result **Parentheses** give 3+2=5 first; multiplying by 5 gives 25. **Parentheses** explicitly group an operation.

Without them, Python multiplication takes priority over addition; the same numerals can therefore describe different procedures.

Changed-case question and hint

- Changed-case practice: Replace the final multiplier 5 with 4 in both **expression**s. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. The first result is 11; the grouped result is 20.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: The first result is 11; the grouped result is 20. 3+2*4=3+8; (3+2)*4=5*4.

Repair and keep the record

Review Values variables and types (python-bridge/2). Goal: Distinguish integer, floating-point, text and Boolean values. Short practice: Predict both print **output**s and explain why they differ.

Does count+tag work, and what happens to each valid **output** after count becomes ten?. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

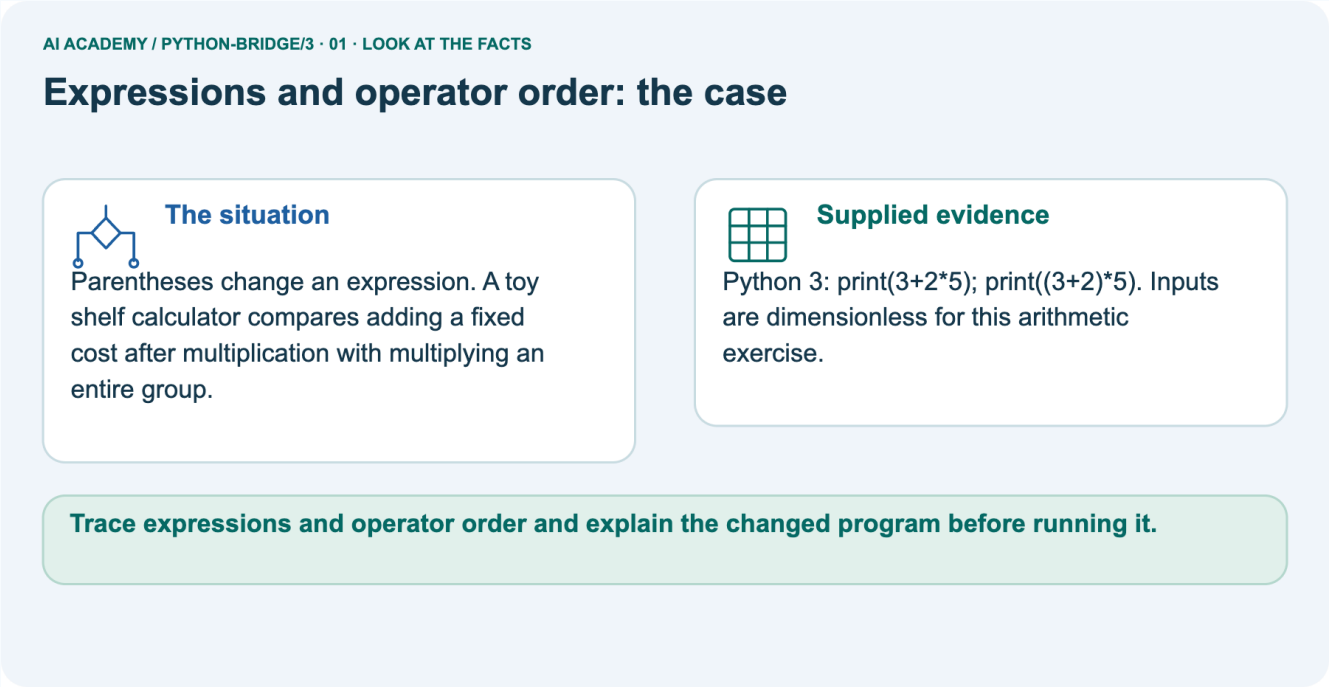
▼ **Reference · The six-picture journey**

REFERENCE · SECTION 18

Reference · The six-picture journey

> Choose the picture needed for the present question.

01. Expressions and operator order: the case



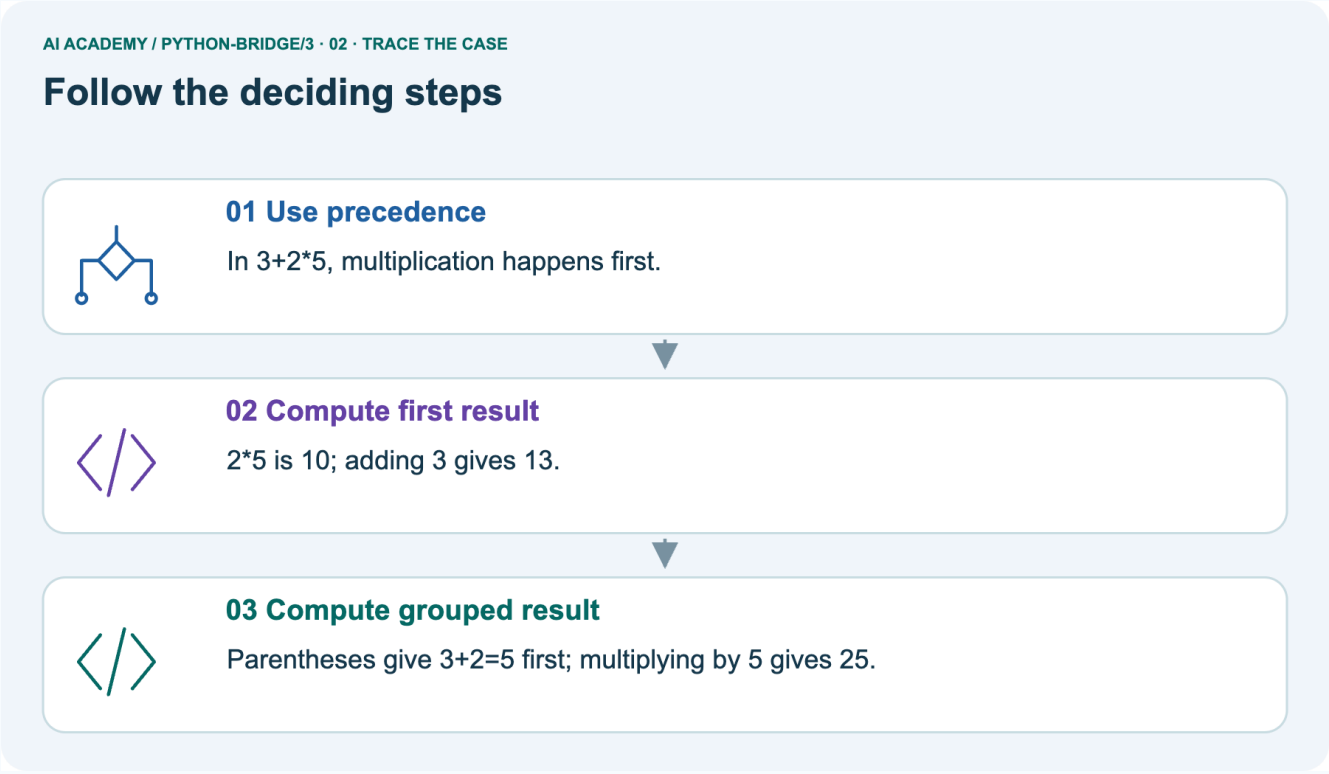
Follow the labelled picture with the matching explanation. Open the vector diagram

Parentheses change an **expression**. A toy shelf calculator compares adding a fixed cost after multiplication with multiplying an entire group. Supplied facts: Python 3: print(3+2*5); print((3+2)*5).

Inputs are dimensionless for this arithmetic exercise.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

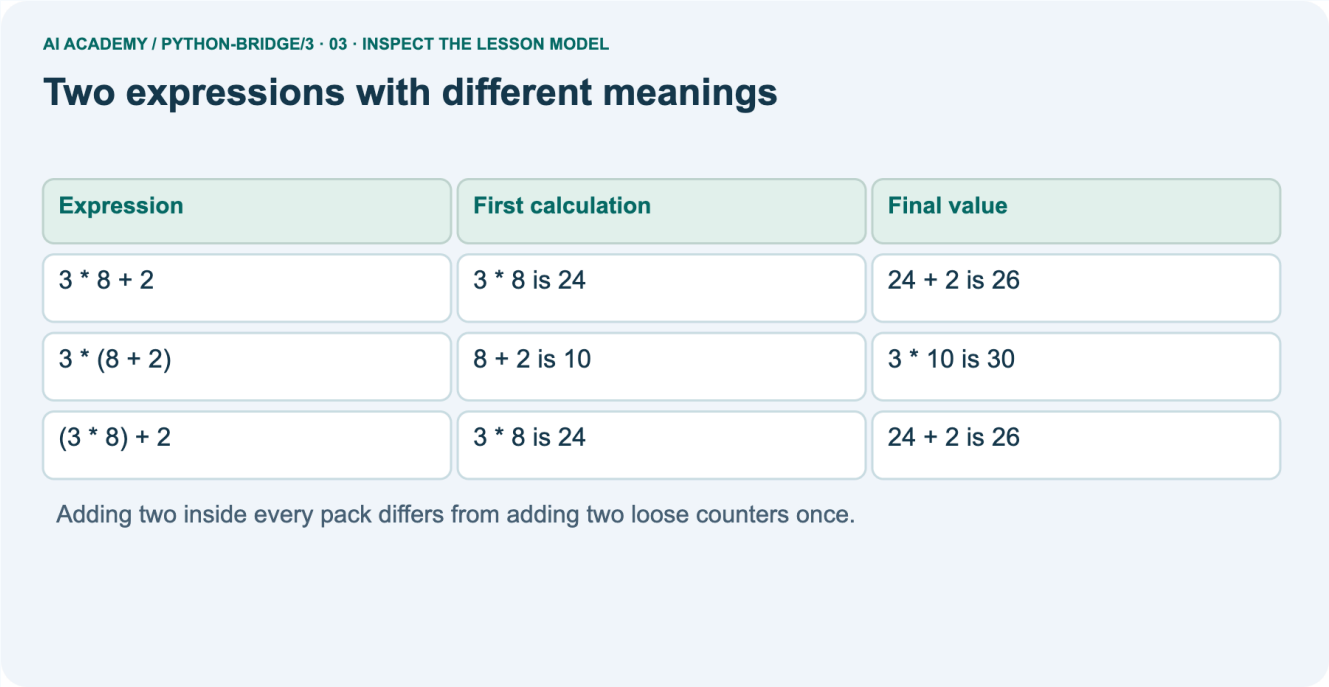


Follow the labelled picture with the matching explanation. Open the vector diagram

Use **precedence**: In 3+2*5, multiplication happens first. → Compute first result: 2*5 is 10; adding 3 gives 13. → Compute grouped result: **Parentheses** give 3+2=5 first; multiplying by 5 gives 25.

Point to each numbered stage and explain which recorded **input** it uses.

03. Two expressions with different meanings

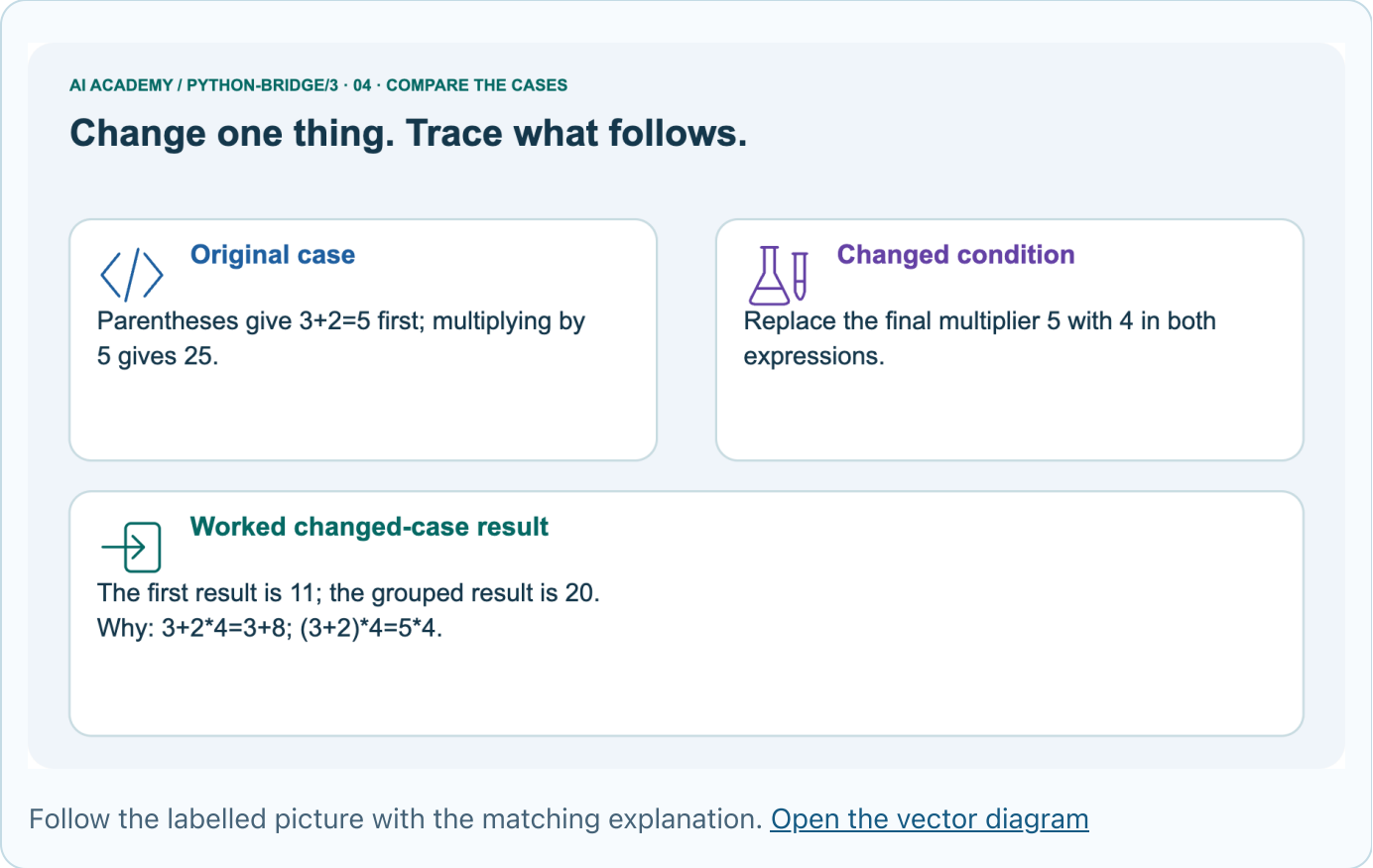


Follow the labelled picture with the matching explanation. Open the vector diagram

Two **expression**s with different meanings. 3 * 8 + 2 | 3 * 8 is 24 | 24 + 2 is 26. 3 * (8 + 2) | 8 + 2 is 10 | 3 * 10 is 30. (3 * 8) + 2 | 3 * 8 is 24 | 24 + 2 is 26

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



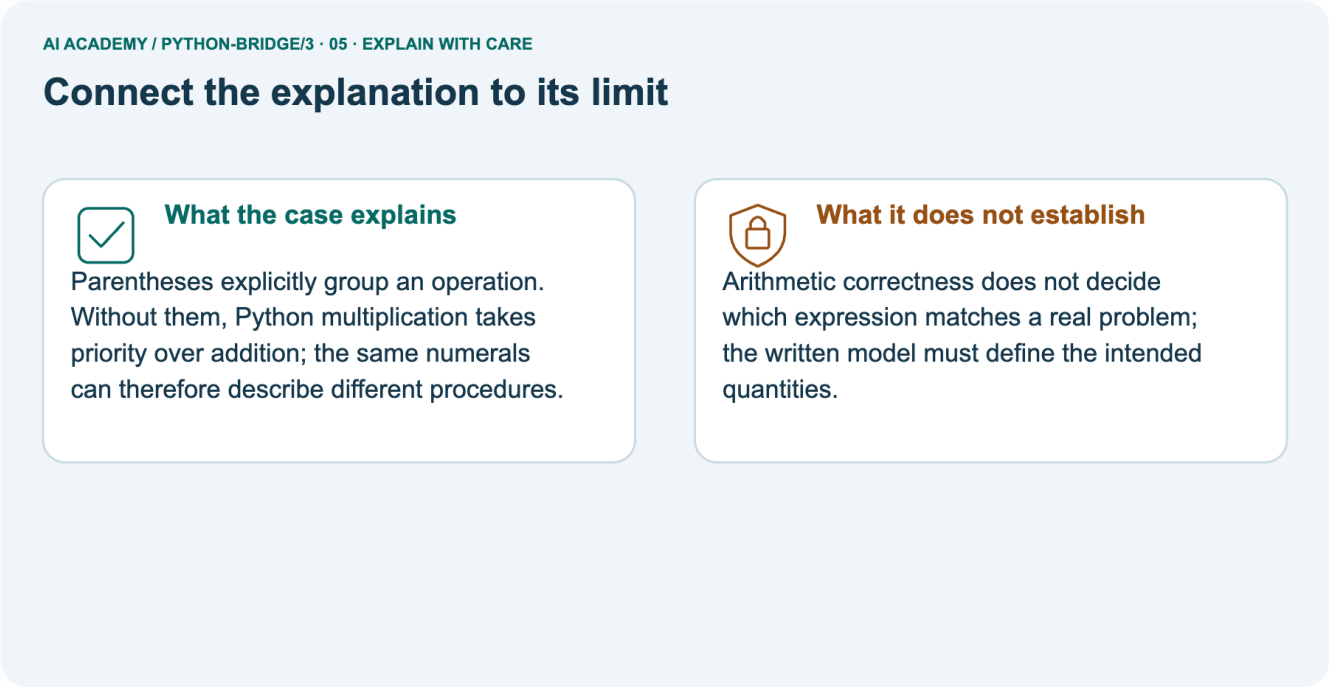
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: **Parentheses** give 3+2=5 first; multiplying by 5 gives 25. Changed condition: Replace the final multiplier 5 with 4 in both **expression**s. Expected result: The first result is 11; the grouped result is 20.

Reason: 3+2*4=3+8; (3+2)*4=5*4.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



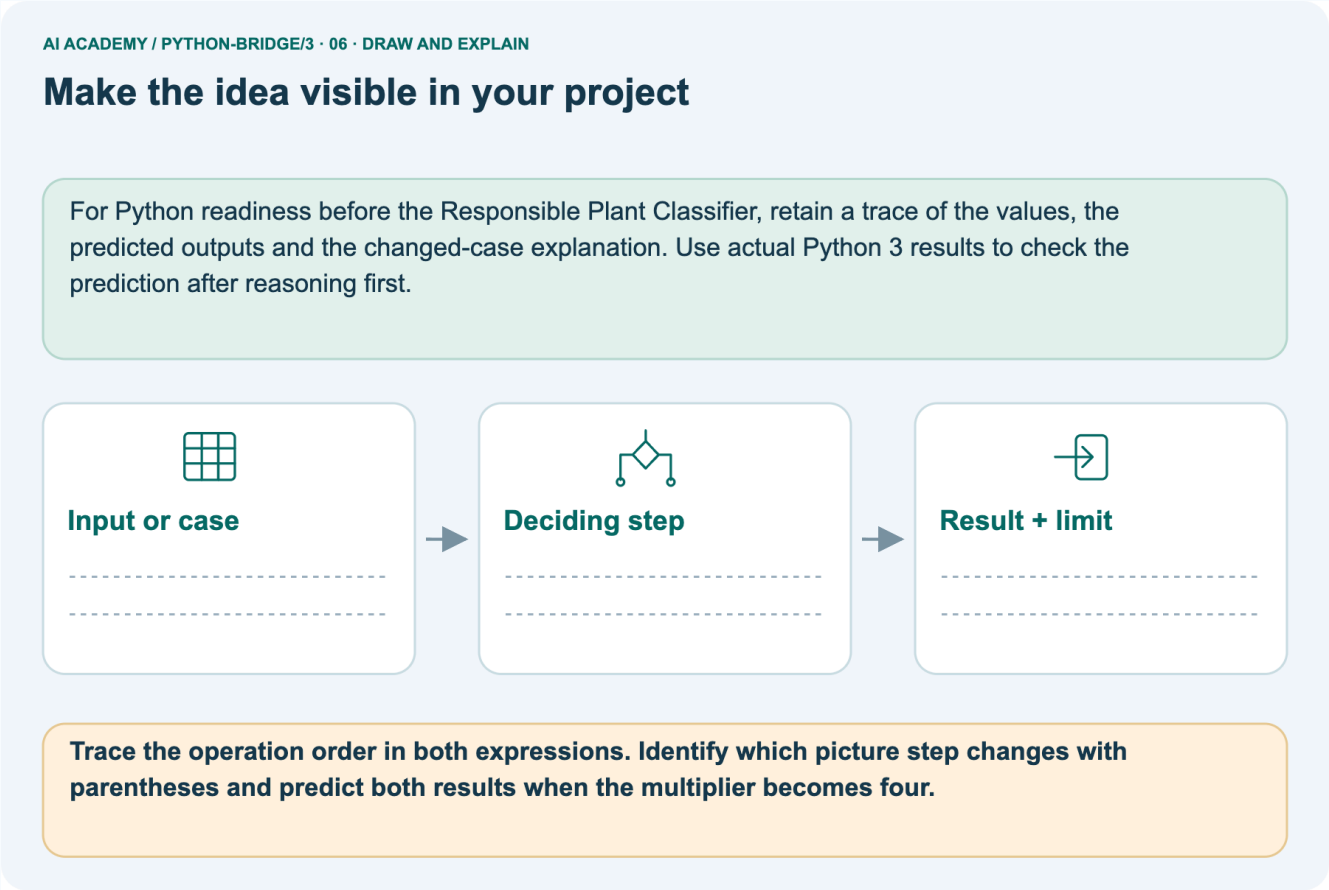
Follow the labelled picture with the matching explanation. Open the vector diagram

Parentheses explicitly group an operation. Without them, Python multiplication takes priority over addition; the same numerals can therefore describe different procedures.

Limit: Arithmetic correctness does not decide which **expression** matches a real problem; the written model must define the intended quantities.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Draw your **input**, deciding step, result and limit. Supported question: Trace the operation order in both **expression**s.

Identify which picture step changes with **parentheses** and predict both results when the multiplier becomes four.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Expressions and operator order: the case

02. Follow the deciding steps

03. Two expressions with different meanings

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.